Expert Systems With Applications 284 (2025) 127952

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

# A Plug-in for cognitive diagnosis method based on correlation representation under long-tailed distribution

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ARTICLE INFO

*Keywords:*
Cognitive diagnosis
Long-tailed
Correlation
Graph representation learning

ABSTRACT

Cognitive diagnosis is a fundamental task in intelligence education, which aims to discover students' proficiency for specific knowledge concepts. Existing cognitive diagnosis models are trained on the basis of sufficient student response records. In applications, however, these records usually follow a long-tailed distribution, i.e. there are only a few students with sufficient records, and a large number of students with a handful of records. The sparsity of records poses a challenge for cognitive diagnosis. To this end, a plug-in based on correlation representation is proposed to address cognitive diagnosis under long-tailed distribution, in which, the correlation representation between head students and tail students is learned to address the sparsity of long-tailed records. In particular, correlation representations are learned in view of both the cognitive state and the learning mode, which are learned based on the node representation and the subgraph representation, respectively. The correlation representation is then used as a plug-in to enhance the representation of long-tailed students and their related exercise and knowledge concepts. With the enhanced representations, the diagnostic performance of tail students is improved. Extensive experiments evaluate the improvement for diagnosis performance and the good compatibility of our plug-in component. Our code is available at https://github.com/joyce99/Wangmian.

## 1. Introduction

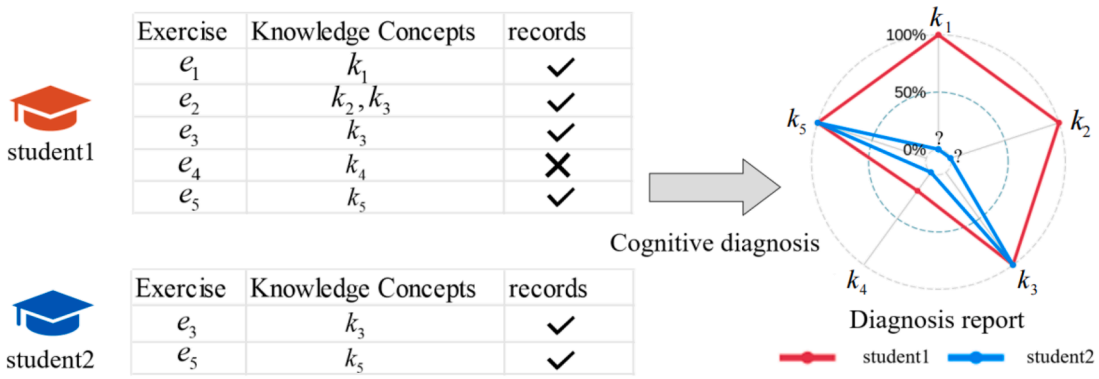
In recent years, the application of artificial intelligence technology in the field of education has attracted widespread attention (Anderson, Huttenlocher, Kleinberg, & Leskovec, 2014a,b). Cognitive diagnosis (CD) is one of the important tasks, and it aims to evaluate students' mastery of knowledge concepts based on student responses to questions, which are also called response records (Cantabella, Martínez-España, Ayuso, Yáñez, & Muñoz, 2019). CD plays an important role for downstream tasks such as teaching evaluation (Adams, Wilson, & Wang, 1997; e Silva et al., 2023) and resource recommendation (Bu et al., 2022; Hu, Liu, & Bu, 2020; Tang & McCalla, 2005).

Established CD methods can be divided mainly into three categories: traditional methods, neural network methods, and graph neural network methods. Traditional methods (Waheed et al., 2023), such as IRT and MIRT (Ackerman, 2014; Embretson & Reise, 2013), model the interaction between students' proficiency and the exercise difficulty from the response records with manually designed interaction functions, making it difficult to capture the complex interaction between students and exercises. The other type is deep learning methods, which use the neural network (NN) or graph neural network (GNN) to learn the representation of students, exercises, and knowledge concepts (Song et al., 2023; Wang et al., 2020) and to model the interaction (Gao et al., 2021; Qi et al., 2023; Wang, Huang, Cai, & Chen, 2021). Although NN methods achieve high power in the fitting and handling of large-scale data sets of interaction functions, structural information is not considered (Shao, Gao, Chen, Yang, Zhang, Wang, 2024). To utilize the structure of data further, graph neural network methods (Gao et al., 2021) treats the students, exercises, and knowledge concepts as a graph, and learns the representation with graph convolution network (GCN). CD methods based on graphs are prone to achieve better performance.

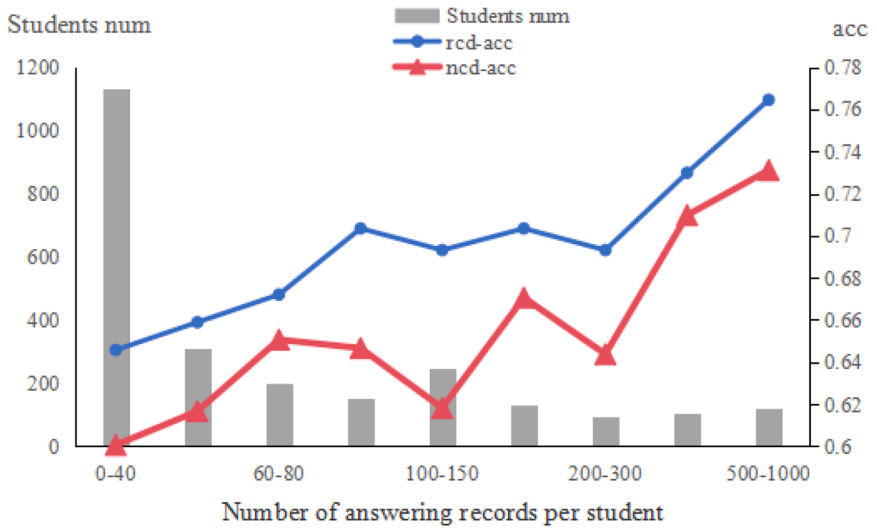
However, the good performance of the above methods depends on sufficient data, including the response records of students on exercises, the association between exercises and knowledge concepts (Chen, Xiangzhi, Wu, Le, Liu, Fei, Chen, Lei, Zhang, Kun, Hong, Richang, Wang, Meng, 2024), and so on. However, in applications, the records of

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https://doi.org/10.1016/j.eswa.2025.127952
Received 14 May 2024; Received in revised form 23 April 2025; Accepted 27 April 2025
Available online 29 April 2025
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(a) Under-diagnosis results of students with fewer records.




(b) Statistics of diagnosis results of long-tailed data.

**Fig. 1.** The challenges of long-tailed distribution in CD.

students answering exercises are often insufficient and usually follow a long-tailed distribution. As shown in Fig. 1a), compared with student 1, student 2 has fewer response records, the sparsity of records makes it more challenging to diagnose the cognitive state of student 2. In addition, we perform the diagnosis performance statistics in the ASSIST 2009–2010 (Heffernan & Heffernan, 2014) dataset, as shown in Fig. 1b). The average number of response records completed by one student is around 107. Among them, students with more than 100 records (called head nodes) account for only 19 %, while students with fewer than 40 records (called tail nodes) account for 45 %. Consequently, existing CD methods obtain significantly lower diagnosis accuracy for tail students than for head students. In particular, the average accuracy of head students with 500–1000 records by NCD (Wang et al., 2020) and RCD (Gao et al., 2021) is 74 %, while the average accuracy of tail students with 0–40 records is 62 %. Obviously, it is challenging for cognitive diagnosis under long-tailed distribution owing to the sparsity of response records.

Fortunately, the long-tailed distribution of response records between exercises and students is beginning to raise concerns. There are a handful of works that focus on this issue. SCD (Wang, Zeng, Yang, & Zhang, 2023) alleviates the long-tailed problem by balancing the degree of the head and tail nodes in the graph, which relatively increases the proportion of records for the tail nodes in the graph learning process. EIRS (Yao et al., 2023) proposes a general auxiliary framework that generates potential response records based on the assumption of cognitive state, i.e. if a student has mastered a knowledge concept, he will obtain accurate response records on those exercises related with this knowledge concept. Although these methods can achieve a certain improvement in diagnostic performance, they still have some shortcomings.

1) Improvement in diagnosis for tail students is at the expense of decreased head students. Some methods reduce the sparsity of tail nodes relatively by sampling the records of head students, which will balance the record number of tail and head students in the representation process with GNN. However, in applications, the records of head students are actually reduced, which will influence the diagnosis of head students. Thus, how to improve the diagnosis of tail students while maintaining the performance of head students is an important issue for CD under long-tailed distribution.

2) The generation of potential records are explored insufficiently and inaccurately. Some methods address the sparsity of records by generating potential records based on the assumption of a cognitive state. The generation can be seen as a prediction of potential records in view of observed records, which are prone to introduce some noisy records and then influence both the head and tail students. However, which exercises students choose to respond is usually influenced by various factors, such as learning habit, and students' majority. The record generation only based on the assumption of cognitive state makes the generated records

insufficient. Thus, finding more accurate potential records is another issue for CD in a long-tailed distribution.

In this paper, a plug-in based on correlation representation is proposed to address the cognitive diagnosis under long-tailed distribution, in which the representation enhancement with noise immunity, rather than data generation, is used to address the sparsity of records. Moreover, the representation enhancement is performed with the correlation in two views, cognitive state and learning mode. Cognitive state correlation refers to the correlation and similarity of two students' cognitive states for a given knowledge concept, which is derived from the cognitive state assumption. If two students have the same or similar cognitive states for some knowledge concepts, one student will respond correctly to the exercises that the other student has responded to. On the other hand, learning mode correlation refers to the correlation and similarity of exercise selection tendency due to factors such as professional background and study habits. A student is more likely to choose the exercises that have been chosen by students who have the same educational background or learning path with him. For two students majoring in computer science, if one student has done exercises related to "data structure" and "operation system", another student is likely to choose the exercises related to "data structure" and "operation system". In this paper, the correlation representation of cognitive state and learning mode are learned based on node representation and sub-graph representation respectively, and both views of correlation representation are used to enhance the representation and improve the diagnosis performance of tail students. Our contributions are summarized as follows.

1. To address the sparsity of records in long-tailed distribution, a plug-in of representation enhancement rather than data generation is proposed to improve the representation and diagnosis performance. The representation enhancement makes CD more robust and does not suffer from sacrificing the performance of the head students.

2. The representation enhancement is performed from two views, cognitive correlation and learning model correlation, which are learned based on the node representation and sub-graph representation, respectively. The enhancement in two views can eliminate the sparsity to a greater extent.

3. Extensive experiments on three benchmark datasets demonstrate that our plug-in component can improve the CD performance of tail students while maintaining that of head students. Additionally, our plug-in can be easily plugged into most CD methods to achieve performance improvements.

## 2. Related work

In this section, CD methods based on NN and GNN are introduced first, and then the research on long-tailed distribution in graphs is introduced.

### 2.1. Cognitive diagnosis methods based on neural network

NCD (Wang et al., 2020) is the first work to use a neural network, consisting of two steps, representation learning and interaction modeling (Zhou et al., 2021). The following work focuses on the improvement in view of the two steps. 1) The step for representation learning. In order to improve the representation of students, ECD (Huang et al., 2021) uses a hierarchical attention network to aggregate student background information and students' historical cognitive state to generate students' context representation, which can improve students' cognitive state. KSCD (Ma et al., 2022) multiplies the embedding of the student and the embedding of the exercise with the embedding of knowledge to learn the student's mastery of knowledge and the corresponding difficulty of knowledge, which enriches the representation of the students and exercise. 2) The step for interaction modeling. DIRT (Cheng et al., 2019) learns students' proficiency and question difficulty by exploiting the interaction between the exercises' texts and the knowledge concepts. Deep-IRT (Yeung, 2019) uses a dynamic key-value memory network to model the interaction between the difficulty of the exercises and the cognitive state of the students. NCD (Wang et al., 2020) models the interaction between students and exercises with multilayer neural networks to replace the manually designed interaction function, which can verify the interpretability through the monotonicity hypothesis.

Although the above methods make full use of the power of neural networks, they treat the students, exercises, and knowledge concepts as three parts independently, ignoring the structural information among students, exercises and concepts, which makes the representation limited.

### 2.2. Cognitive diagnosis methods based on graph neural network

The remarkable efficiency of GNN (Zhou et al., 2020) has attracted considerable attention in various domains. For cognitive diagnosis, students, exercises, and knowledge concepts form a structured graph (Zhang, Song, Huang, Swami, & Chawla, 2019), making graph-based cognitive diagnosis a focal point of research.

GKT-CD (Zhang, Mo, Chen, & He, 2021) uses graph structure and employs a gated GNN to model the students' learning path, inferring their cognitive states and enhancing the diagnosis accuracy. RCD (Gao et al., 2021) establishes a hierarchical graph comprising three local graphs, student-exercise, exercise-knowledge concept, and knowledge concept dependency graphs, to learn the representations of students, exercises, and knowledge concepts with GAT, which improves the efficiency of cognitive diagnosis. MRCD (Wu et al., 2022) addresses variances in student-exercise interactions by partitioning student response records into correct and incorrect interaction graphs and uses a contrastive learning scheme with relational data augmentation to alleviate the sparsity arising from graph partitioning.

Although these methods enhance node representations to improve diagnosis accuracy by using graph structures, they overlook the long-tailed problem inherent in graphs.

### 2.3. Cognitive diagnosis methods for the long-tailed distribution

The long-tailed problem is prevalent in applications and has garnered significant research attention. Recent studies such as metatail2vec (Liu, Zhang, Fang, Zhang, & Hoi, 2020) and Tail-GNN (Liu, Nguyen, & Fang, 2021b) are proposed to tackle long-tailed challenges in social network analysis (Freeman et al., 2004; Marin & Wellman, 2011), recommendation systems (Anneroth, Batsakis, & Luna, 1987; Lika, Kolomvatsos, & Hadjiefthymiades, 2014), and bioinformatics (Yi, You, Huang, & Kwoh, 2022).

There are also a few works that focus on long-tailed problem in CD. SCD (Misra & van der, 2020) prunes associations between certain students and exercises based on the degree of the student in the process of self-supervised learning. This strategy makes the model pay more attention to long-tailed students. However, it must balance the number of records between head and tail students, compromising the diagnosis accuracy of head students. EIRS (Yao et al., 2023) addresses the long-tailed challenge by generating potential records based on mastered knowledge concepts, thus increasing the number of tail students' records. However, potential records may be noisy.

Although these methods strive to balance answering records between head and tail students, they address the long-tailed problem at the data level, which makes the improvement of tail students limited.

## 3. Proposed method

### 3.1. Problem definition

In a teaching system, there are $N$ students, $M$ exercises, and $K$ knowledge concepts, which are denoted as $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$, $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$, and $\mathcal{K} = \{k_1, k_2, \ldots, k_K\}$, respectively. Each student

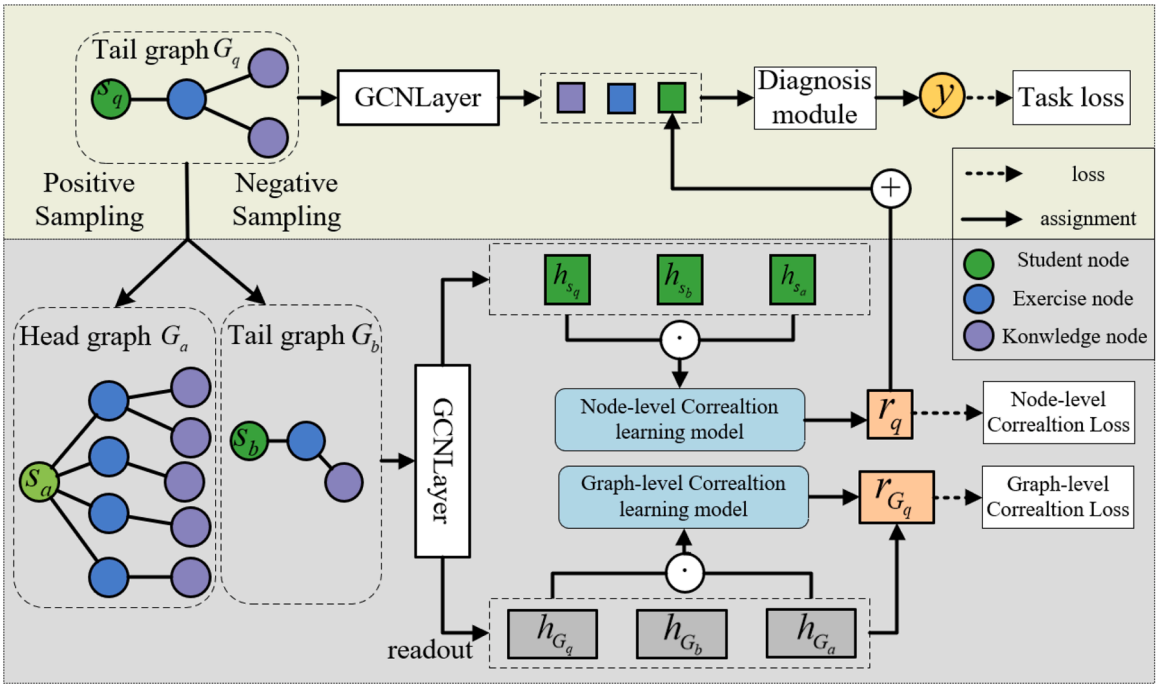



**Fig. 2.** framework of HT-GCD.

selects some exercises to practice and their response records manifest themselves as triples, recorded as $\mathcal{R} = (s_i, e_j, r_{ij})$, where $r_{ij}$ denotes the score (usually normalized from 0 to 1) of the student $s_i$ on exercise $e_j$. Additionally, there is an exercise-concept correlation matrix $Q = {q_{ij}}^{M\times K}$ with binary elements $q_{ij} = 1$, which indicates exercise $e_i$ contains or is related to concept $k_j$. Generally, $Q$ is annotated by experts annually. In this paper, we focus on the long-tailed distribution of records, which denotes a large portion of students with limited response records on exercises. Our task is to mine the proficiency of both head and tail students for each concept and predict their performance in doing the related exercises.

In this paper, a representation enhancement is proposed rather than a data augmentation to eliminate the sparsity of records. The representation enhancement is based on the correlation representation, which is learned between head students and tail students in view of cognitive state and learning mode. Our proposed representation enhancement is treated as a plug-in component of existing methods, as shown in Fig. 2. The plugged method includes three modules: 1) Node representation. The representation of graph nodes, such as students, exercises, and knowledge concepts, is learned with GCN. This module is similar to other CD methods. 2) Learning of correlation representations. For each tail node, the correlation representation is learned based on its akin head nodes in view of cognitive state and learning mode, which are represented based on node and sub-graph respectively. 3) Correlation-fused representation and diagnosis. The correlation representation is fused to enhance the representation of tail students, then the diagnosis model is trained.

### 3.2. Node representation

In this subsection, the embeddings of students, exercises, and knowledge concepts are learned with GCN. This step is based on the graph and the embeddings of three types of nodes are learned with GCN.

Students, exercises, and knowledge concepts form a graph. And GCN (Yao, Mao, & Luo, 2019) is used to represent the nodes by aggregating its neighboring nodes. The student embedding is learned by aggregating the exercise embedding, the exercise embedding aggregates the student embedding and the knowledge embedding. The aggregation process is as follows.

$$h_{s_i}^{(l+1)} = \sigma\left(\sum_{j\in N_{s_i}^{e}} W_{ij}^{(l)} h_{e_j}^{(l)} + W_0^{(l)} h_{s_i}^{(l)}\right) \tag{1}$$

$$h_{e_j}^{(l+1)} = \sigma\left(\sum_{k\in N_{e_j}^{c}} \sum_{j\in N_{e_j}^{s}} W_{ij}^{(l)} h_{s_i}^{(l)} + W_{jk}^{(l)} h_{c_k}^{(l)} + W_0^{(l)} h_{e_j}^{(l)}\right) \tag{2}$$

$$h_{c_k}^{(l+1)} = \sigma\left(\sum_{j\in N_{c_k}^{e}} W_{kj}^{(l)} h_{e_j}^{(l)} + W_0^{(l)} h_{c_k}^{(l)}\right) \tag{3}$$

where $h_{s_i}^{(l)}$, $h_{e_j}^{(l)}$, and $h_{c_k}^{(l)}$ represent the embedding of students, exercises, and knowledge concepts at layer $l$ of GCN, $W^{(l)}$ are trainable parameter matrices. The dimension of these embeddings is generally set to the number of knowledge concepts $K$ in the experiment.

### 3.3. Correlation representation learning

In this subsection, head and tail students are separated first, and then a set of head students similar to each tail student is identified. Subsequently, the correlation representation between head and tail students is learned in terms of cognitive state and learning mode, respectively.

#### 3.3.1. Sampling head students similar to tail students

Firstly, we need to separate the head and tail students. In particular, the students are ranked according to the number of exercises they responded to. The top 20 % of students constitute the head nodes $H_S$, while the remaining 80 % of students form the tail nodes $T_S$.

Secondly, we find similar head nodes for each tail node $s_q \in T_S$, assuming that the similarity between students indicates the correlation representation. The similar candidate head nodes are those nodes that have a high similarity to the tail node $s_q$. Here, the similarity is computed with the simple matching coefficient (SMC) (Gower, 1971) between the tail $s_q \in T_S$ and the head nodes $s_a \in H_S$, as follows.

$$Similarity(s_a, s_q) = \frac{R_{s_a e} \cap R_{s_q e}}{R_{s_q e} \cap \sum_{i=1}^{S} R_{s_i e}} \tag{4}$$

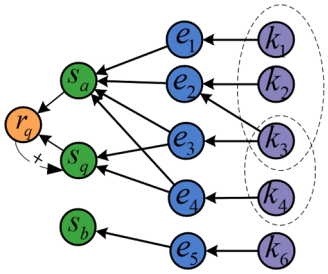


**Fig. 3.** The illustration of Cognitive correlation representation. $s_q$ is a tail student, $s_a$ is a head student similar to $s_q$, and $s_b$ is a tail student dissimilar to $s_q$. The representation of students reflects the cognitive state on knowledge concepts, which aggregates some knowledge concepts. Then, $r_q$ is generated by the correlation representation between $s_a$ and $s_q$, as shown with solid arrows, and then $r_q$ is added to $s_q$ to enhance the cognitive representation of tail student, as shown with dashed arrows.

where $R_{(s_a e)}$ and $R_{(s_q e)}$ denote the set of response records for head student $s_a$ and tail student $s_q$. Here only the correct response are considered.

With Eq. (4), the similarity matrix $W \in R^{|SH|*|ST|}$ is computed, which makes up the similarity coefficient matrix. If $w_{aq}$ exceeds the similarity threshold, the head student $s_a$ is deemed to be correlated to the tail $s_q$. Here, the threshold is set as the mean of the $q$th row of the $W$ matrix, denoted as $AVERAGE(w_q)$. For the tail student $s_q$, the set of similar candidate head nodes is denoted $H_q = \{s_a | w_{aq} > \alpha \times AVERAGE(w_q)\}$, where $\alpha$ serves as a manually adjusted coefficient.

### 3.3.2. Cognitive correlation representation

We learn the cognitive correlation representation between head and tail students based on the node embeddings. Student embedding is closely related to proficiency, which contains the interaction between students and knowledge concepts. Thus, it reflects the cognitive state of the students on concepts to some extent.

If two students correctly answer exercises related to the same knowledge concept, it is generally regarded that they have mastered these knowledge concepts and their cognitive states are correlated. As illustrated in Fig. 3, the tail student $s_q$ masters the concept of knowledge $k_3$, while the head student $s_a$ masters $k_1$, $k_2$, and $k_3$. Consequently, the correlation representation of the cognitive state $r_q$ is learned and transferred from $s_a$ to $s_q$ to enrich the representation of the tail student $s_q$. Then the tail student $s_q$ is deemed to possess potential mastery over $k_1$ and $k_2$.

A correlation learning function $\psi$ comprising a neural network is designed to learn the cognitive correlation $r_q$ between the tail student $s_q$ and its corresponding head student $s_a$. The cognitive correlation representation is learned as follows.

$$r_q = \psi(h_{s_q}^{(l)}, h_{s_a}^{(l)}) = LeakyRelu\left(A_q\left(h_{s_q}^{(l)} \odot h_{s_a}^{(l)}\right)B_q\right)_{s_a \in H_q} \tag{5}$$

where $A_q \in R^K$ is the cognitive correlation weight vector and $B_q \in R^{K*K}$ is the cognitive correlation transfer matrix. The cognitive correlation weight vector is used to adjust the proportion of the information of $h_{s_q}^{(l)}$ in $r_q$, the cognitive correlation transfer matrix is used to adjust the proportion of $h_{s_a}^{(l)}$ in the representation of cognitive correlation.

Furthermore, the correlation representation $r_q$ is guided by driving $r_q$ closer to its similar head student $s_a$ and farther away from the other tail student $s_b$. Hence, the constraint loss is defined as follows.

$$L_{rel}^{node} = -\sum_{s_a \in H_q, s_b \in T_S, s_b \neq s_q} \ln \sigma(sim(r_q, h_{s_a}) - sim(r_q, h_{s_b})) \tag{6}$$

where $s_q$, $s_a$, $s_b$ belong to the student node set $S_{node}$. $(s_q, s_a)$ forms positive pairs, in which $s_a$ is similar with current tail node $s_q$, while $(s_q, s_b)$ forms negative pairs, in which $s_b$ is a different tail node from current tail node $s_q$. $sim$ is the similarity function, and we adopt the cosine function here. The sigmoid function $\sigma(x) = \frac{1}{(1+e^{-x})}$ is adopted as an activity function.

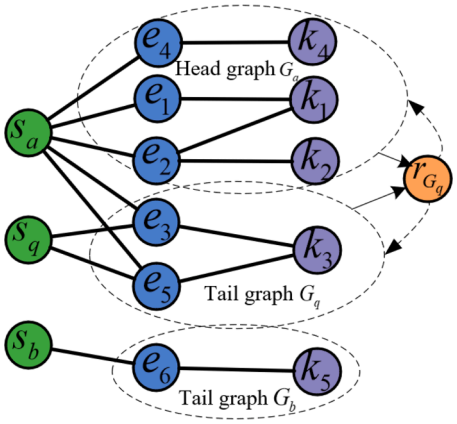



**Fig. 4.** The illustration of learning mode correlation. The correlation of learning mode $r_{G_q}$ is learned from the graph representations of subgraph $G_a$ and $G_q$, as shown with solid arrows. And then $r_{G_q}$ is treated as the constraint to in turn optimize the representations of the nodes in the subgraphs during the backpropagation process, as shown with the dashed arrows.

### 3.3.3. Learning mode correlation representation

In addition to cognitive correlation, we also learn the correlation of learning mode, which implies that two students have some correlation in their inclination to choose exercises and concepts. Here, the learning mode of students are modeled using a graph-level representation that includes nodes of exercises and knowledge concepts.

If two students have the same tendency to select exercises and knowledge concepts, there exists a correlation between the learning modes between them. In practice, the learning mode can be influenced by factors such as their majority and some habits. As illustrated in Fig. 4, the tail student $s_q$ selects two exercises $e_3$ and $e_5$, which are related to the concept $k_3$, while the head student $s_a$ selects exercises $e_1, e_2, e_3, e_4, e_5$. Consequently, there exists similarity or correlation in the selection of exercises between $s_q$ and $s_a$. Thus, the correlation representation of the learning mode $r_{G_q}$ is learned and transferred from $s_a$ to $s_q$. And then the tail student $s_q$ is deemed to possess a potential response to exercises such as $e_1$, $e_2$ and $e_4$. And the exercise-knowledge sub-graph implies the tendency of student choosing exercises, which reflects the learning mode of students to a certain extent.

The correlation of learning modes between $s_q$ and $s_a$ can be captured with the related subgraph of the students. Firstly, we design a learning mode correlation function $\phi$ based on sub-graph representations $G_q$ and $G_a$ to extract the correlation of learning modes between head and tail students.

$$r_{G_q} = \phi(h_{G_q}^{(l)}, h_{G_a}^{(l)}) = LeakyRelu\left(C_q\left(h_{G_q}^{(l)} \odot h_{G_a}^{(l)}\right)D_q\right)_{s_a \in H_q} \tag{7}$$

$$h_{Gi} = READOUT(\{h_{Ei}, h_{Ki}\}) \tag{8}$$

where $G_q$ and $G_a$ represent the subgraph of student $s_q$ and $s_a$, and $G_q$ is a graph containing the responded exercises and knowledge concept. The representation of $G_q$ and $G_a$ are learned with READOUT function (Gholamalinezhad & Khosravi, 2020). $C_q \in R^K$ is the learning mode weight vector, and is used to adjust the proportion of the information of $h_{G_q}^{(l)}$ in $r_{G_q}$. $D_q \in R^{K*K}$ is the learning mode transfer matrix, and is used to adjust the proportion of $h_{G_a}^{(l)}$ in the representation of learning mode correlation.

Secondly, the learning mode of the tail student $G_q$ is driven close to its similar head subgraph $G_a$ and farther away from another tail subgraph $G_b$, enhancing the representation of exercises and concepts in the

subgraph of $G_q$.

$$L_{rel}^{subg} = -\sum_{s_a \in H_q, s_b \in T_S, s_b \neq s_q} \ln \sigma(sim(r_{Gq}, h_{Ga}) - sim(r_{Gq}, h_{Gb}))$$

Considering that this correlation of the learning mode is based on the representation of the subgraphs, including the exercises and knowledge concepts, we treat $r_{G_q}$ as a constraint of the representation learning of exercises and knowledge concepts, not the enhancement of the student representation directly.

### 3.4. Cognitive diagnosis based on enhanced representation

In this subsection, the correlation representation is plugged in to the CD methods to improve performance, especially for the tail students.

Firstly, the correlation representation is used to enhance the representation of tail students. For each tail student $s_i$, we enrich its original embedding by incorporating correlation representation $r_i$. Here, we directly combine the initial student representations $h_{s_i}$ and the correlation representation $r_i$ with a second-order operation, as follows:

$$h'_{s_i} = h_{s_i} \oplus r_{s_i} \tag{9}$$

where $\oplus$ denotes the second-order operation. Some operations such as addition, subtraction, and product of vectors, and neural networks with fully-connected layers can be used. And then the enhanced representation $h'_{s_i}$ is further utilized for the cognitive diagnosis of the students.

Secondly, the diagnosis model is trained based on the enhanced representations, and model training is performed using the classification loss of predicted results. Similarly to RCD, a fully connected layer $F_s$ is used to learn student proficiency ($h_{pro}^{s_i}$) and difficulty in exercise ($h_{diff}^{e_j}$) based on the embeddings of the students and exercises.

$$h_{pro}^{s_i} = sigmoid(F_s(h'_{s_i})) \tag{10}$$

$$h_{diff}^{e_j} = sigmoid(F_s(h_{e_j})) \tag{11}$$

The interaction is modeled with a linear layer $F$. To verify the correctness of the diagnosis, a prediction function is employed to predict the student's score:

$$y_{ij} = \frac{1}{|N_{e_j}^k|} \sum_{k \in N_{e_j}^k} \sigma(F(h_{prof}^{s_i} - h_{diff}^{e_j})) \tag{12}$$

where $y_{ij}$ denotes the result of student $s_i$ in answering exercise $e_j$. $N_{e_j}^k$ represents the set of knowledge concepts involved in exercise $e_j$. $|\cdot|$ is the counting symbol.

To train the model, the loss function for the primary task is the regular loss of cross-entropy between the predicted score $y$ and the true score $r_{ij}$.

$$L_{task} = -\sum_{i,j} \left( r_{ij} log y_{ij} + (1 - r_{ij}) \log(1 - y_{ij}) \right) \tag{13}$$

Thus, the overall goal is formulated as follows:

$$L = L_{task} + \mu L_{rel}^{node} + (1 - \mu) L_{rel}^{subg} \tag{14}$$

where $\mu$ is a trade-off parameter adjusting the weight of each component, and its value is set to 0.5.

### 3.5. The analysis of proposed method

Our proposed method is a plug-in that can be applied to different base methods with good compatibility. When applied to NN-based methods, such as NCD and KSCD, only the cognitive correlation $r_q$ can be integrated. While when applied to graph-based methods such as RCD, SCD, etc., both cognitive correlation $r_q$ and learning mode correlation $r_{G_q}$ can be used.

In addition, we analyze the time cost of our method. It includes the calculation of two correlations, $r_q$ and $r_{G_q}$ for each tail student. The computation of $r_q$ is shown in Eq. (5), which performs the matrix product operation of the student representation and the cognitive correlation weight matrix (with a time complexity of $O(K^2 \cdot h_s)$, where $K$ is the number of knowledge concepts) and then performs LeakyRelu activation on the product result (with a time complexity of $O(h_s)$). Therefore, the computational time complexity of $r_q$ is $O(K^2 \cdot h_s + h_s)$. Compared to $r_q$, the computation of $r_{G_q}$ requires an additional readout function to obtain the graph representation, where the time complexity of the linear layer is $O(K^2)$. Therefore, the time complexity of $r_{G_q}$ is $O(K^2 + K^2 \cdot h_s + h_s) = O(K^2 \cdot h_s + h_s)$. In general, the time complexity of our plug-in is $O(N * (K^2 \cdot h_s + h_s))$, where $N$ is the number of students.

**Table 1**
Statistics of data sets.

| Statistics | Mooper | ASSIST | JunYi |
|---|---|---|---|
| #Students | 1001 | 2493 | 10,000 |
| #Exercises | 7218 | 17,746 | 835 |
| #Concepts | 3271 | 123 | 835 |
| #Responses | 31,581 | 267,415 | 353,835 |
| #Concepts per exercise | 3.7 | 1.2 | 1 |
| #Responses per student | 15.4 | 107.27 | 35.38 |
| #Tail nodes of student | 531 | 1133 | 6217 |

## 4. Experiment

We conducted extensive experiments on three real-world datasets to demonstrate the effectiveness of our method under a long-tailed distribution.

### 4.1. Dataset and baselines

**Datasets.** Three popular datasets are used and their statistical information is presented in Table 1.

ASSIST (ASSISTments 2009–2010 Skill Builder).[1] A dataset collected by ASSISTments (Online Intelligent Tutor System), which includes student response records from the 2009–2010 school year along with the relationship between included exercises and knowledge concepts. Students with less than 4 records are categorized as tail students, while the remaining students are considered head students.

Mooper (Massive Open Online Practice, MOOP).[2] A large-scale online open practice dataset contributed by the Key Laboratory of Information Systems Engineering of the National University of Defense Technology (Liu et al., 2021a). Mooper originates from large-scale open online practice, comprising online practice data from Touge platform users from 2018 to 2019. Due to the extensive data, many exercises lack the corresponding knowledge associations. Therefore, only question records containing knowledge concepts are selected for experiments, and students with less than 10 records are defined as tail students.

JunYi.[3] Derived from the online learning platform JunYi Academy. This dataset gathers response records from October 2012 to January 2015. Each exercise contains only one knowledge concept, and it provides relationships between knowledge concepts marked by experts, and students with fewer than 10 records are defined as tail students.

**Baselines.** To verify the effectiveness of our method, we select four neural network models NCD, KSCD, RCD, SCD and EIRS as baselines, in which SCD and EIRS contain the mechanism for long-tailed distribution.

NCD (Wang et al., 2020) is one of the first CDMs based on neural networks. It employs multiple neural layers to model learners and exercises, and applies the monotonicity assumption to ensure model interpretability.

[1] https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/skill-builder-data-2009-2010

[2] https://www.educoder.net/

[3] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=1198

**Table 2**
Performance of our correlation representation when plugged in baselines.

| Datasets | ASSIST | | | | MOOPer-80 | | | | Junyi | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | ACC | RMSE | ACC-H | ACC-T | ACC | RMSE | ACC-H | ACC-T | ACC | RMSE | ACC-H | ACC-T |
| NCD | 0.711 | 0.436 | 0.738 | 0.692 | 0.698 | 0.442 | 0.725 | 0.679 | 0.784 | 0.411 | 0.819 | 0.760 |
| NCD-$r_q$ | 0.732 | 0.429 | 0.734 | 0.728 | 0.709 | 0.421 | 0.716 | 0.702 | 0.794 | 0.405 | 0.813 | 0.778 |
| KSCD | 0.715 | 0.432 | 0.746 | 0.694 | 0.735 | 0.432 | 0.766 | 0.713 | 0.815 | 0.396 | 0.849 | 0.791 |
| KSCD-$r_q$ | 0.732 | 0.421 | 0.740 | 0.728 | 0.752 | 0.425 | 0.763 | 0.734 | 0.827 | 0.391 | 0.841 | 0.807 |
| RCD | 0.721 | 0.425 | 0.749 | 0.702 | 0.699 | 0.453 | 0.733 | 0.676 | 0.811 | 0.401 | 0.836 | 0.794 |
| RCD-$r_q$ | 0.732 | 0.421 | 0.742 | 0.728 | 0.722 | 0.439 | 0.747 | 0.709 | 0.824 | 0.393 | 0.836 | 0.812 |
| RCD-$r_q r_{G_q}$ | 0.732 | 0.418 | 0.743 | 0.728 | 0.723 | 0.436 | 0.746 | 0.712 | 0.825 | 0.394 | 0.837 | 0.814 |
| SCD* | 0.721 | 0.436 | 0.738 | 0.706 | 0.715 | 0.436 | 0.752 | 0.705 | 0.786 | 0.423 | 0.823 | 0.713 |
| SCD*-$r_q$ | 0.735 | 0.423 | 0.741 | 0.730 | 0.769 | 0.415 | 0.787 | 0.729 | 0.811 | 0.392 | 0.823 | 0.807 |
| SCD*-$r_q r_{G_q}$ | 0.739 | 0.422 | 0.741 | 0.737 | 0.771 | 0.412 | 0.776 | 0.765 | 0.813 | 0.389 | 0.822 | 0.810 |

KSCD (Ma et al., 2022) uses students' responses to record the relationship between learning knowledge concepts, we obtain the embedded representation of knowledge concept vectors based on neural networks. Then it fuses students and knowledge concept vectors to derive the cognitive status of students.

RCD (Gao et al., 2021) represents learners, exercises, and concepts as nodes in three local relationship graphs, building a multilayer attention network to aggregate the relationships between the nodes in the graph and the relationships between graphs.

SCD (Wang et al., 2023) uses graph neural networks to learn node representations and uses self-supervised learning to improve long-tailed student representations.

EIRS (Yao et al., 2023) is based on the assumption that students are correct in solving exercises on knowledge concepts that they have mastered, and generates potential interaction records of students to alleviate the long-tailed problem. It should be noted that EIRS can also be treated as a plug-in that addresses a long-tailed problem. In its original article, it was combined with multiple baselines.

### 4.2. Experimental details

We set the vector dimensions of students, exercises, and knowledge concepts to $K$ dimensions, where $K$ is the number of knowledge concepts. The GCN layer in Eqs. (1)–(3) is stacked for 2 layers and the Adam optimizer is used to optimize the model parameters with a learning rate of 0.001. The training process runs for 10 epochs. We employ 5-fold cross-validation method, with 60 % of records as training and 40 % as test. Considering that the exercises in the dataset used in three datasets are objective exercises, representing incorrect or correct answers with scores of 0 or 1 respectively, we employ RMSE (root mean square error) and ACC (prediction accuracy) to evaluate the performance of CD.

### 4.3. Overall performance

Table 2 shows a comparative analysis of the effectiveness of our plug-in based on correlation representation with multiple baselines on three datasets. Since NCD and KSCD are not based on a graph, they only incorporate the cognitive correlation at node level (referred to as NCD-$r_q$ and KSCD-$r_q$). In contrast, RCD and SCD are graph-based, both node-level cognitive correlation and subgraph-level learning mode correlations can be plugged (referred to as RCD-$r_q r_{G_q}$ and KSCD-$r_q r_{G_q}$). SCD* is the SCD model that ignores the contrastive learning module. In Table 2, ACC denotes the accuracy of the diagnosis of all students, while ACC-H and ACC-T signify the accuracy of head students and tail students, respectively.

1) For all baselines, the ACC and ACC-H of all students surpass the ACC-T. ACC-H is higher than ACC-T by 4.8 % on average. It highlights the challenge of CD under long-tailed distribution, especially for tail students.

2) Our correlation plug-in can improve the performance of CD under long-tailed distribution. Both ACC and ACC-T witness substantial improvements when our correlation representation is plugged. Compared to NCD, NCD-$r_q$ increases ACC-T by 3.7 %. Similarly, compared to RCD and SCD, RCD-$r_q r_{G_q}$ and SCD-$r_q r_{G_q}$ lead to an improvement of 2.7 % and 6.3 % in ACC-T, respectively. These indicate that our correlation representation can find the right head students to enhance the representation of tail students and then improve the diagnosis performance.

**Table 3**
Runtime(s) of our correlation representation when plugged in baselines.

| Model | ASSIST | MOOPer-80 | Junyi |
|---|---|---|---|
| NCD | 14.15 | 4.24 | 52.79 |
| NCD-$r_q$ | 16.37 | 14.35 | 303.23 |
| KSCD | 16.58 | 19.46 | 1150.32 |
| KSCD-$r_q$ | 31.07 | 30.67 | 2234.16 |
| RCD | 16.98 | 13.83 | 1195.70 |
| RCD-$r_q$ | 31.75 | 21.85 | 2319.66 |
| RCD-$r_q r_{G_q}$ | 33.34 | 22.94 | 2435.64 |
| SCD* | 8396.05 | 2283.89 | 16795.83 |
| SCD*-$r_q$ | 87238.42 | 2980.3 | 180978.33 |
| SCD*-$r_q r_{G_q}$ | 87708.09 | 3066.16 | 181086.61 |

3) The performance of head students is maintained while that of tail students is improved. Compared with baselines, the ACC-H of -$r_q$ and -$r_q r_{G_q}$ improve by [-1.4 %, 3.5 %] on all data sets, with comparable performance on average. It indicates that our correlation representation improves the diagnosis of tail students without compromising head students. This is because the learning and enhancement process of correlation representation rarely influences the representation of head students.

4) Our correlation representation can be plugged in multiple baselines and achieve improvements on three datasets, which shows the effectiveness of correlation representation and the good portability.

Table 3 shows the time cost of our method when applied with other CD models. Our proposed plug-in is combined with other CD models to achieve the diagnostic results. Compared with general CD models, it requires extra time to solve the correlation between the head and tail students. Among three datasets, the large time overhead on the Junyi dataset is mainly due to its large number of tail students. It is worth mentioning that the theoretical analysis shows that the time complexity is affected by the number of knowledge concepts, but the actual time overhead caused by matrix operations is relatively small due to the highly optimized and cache-friendly practical implementation. Therefore, the performance in the experiment is mainly affected by the number of students in the tail students.

### 4.4. Ablation study

#### 4.4.1. Comparison of other mechanisms for long-tail.

Table 4 illustrates the performance comparison between our correlation representation and other long-tailed strategies. -EIRS refers to the Practice Perceptual Information Sampling framework proposed by

**Table 4**
Comparison of different processing for long-tailed distribution.

| Datasets | ASSIST | | | | MOOPer | | | | Junyi | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | ACC | RMSE | ACC-H | ACC-T | ACC | RMSE | ACC-H | ACC-T | ACC | RMSE | ACC-H | ACC-T |
| NCD | 0.711 | 0.436 | 0.738 | 0.692 | 0.698 | 0.442 | 0.725 | 0.679 | 0.784 | 0.411 | 0.799 | 0.724 |
| NCD-EIRS | 0.710 | 0.448 | 0.709 | 0.715 | 0.704 | 0.429 | 0.731 | 0.689 | 0.781 | 0.419 | 0.787 | 0.732 |
| NCD-$r_q$ | 0.732 | 0.429 | 0.734 | 0.728 | 0.709 | 0.421 | 0.716 | 0.702 | 0.794 | 0.405 | 0.813 | 0.778 |
| KSCD | 0.715 | 0.432 | 0.746 | 0.694 | 0.735 | 0.432 | 0.766 | 0.713 | 0.815 | 0.396 | 0.862 | 0.779 |
| KSCD-EIRS | 0.712 | 0.436 | 0.733 | 0.699 | 0.736 | 0.431 | 0.758 | 0.721 | 0.816 | 0.399 | 0.849 | 0.791 |
| KSCD-$r_q$ | 0.732 | 0.421 | 0.740 | 0.728 | 0.752 | 0.425 | 0.763 | 0.734 | 0.827 | 0.391 | 0.841 | 0.807 |
| RCD | 0.721 | 0.425 | 0.749 | 0.702 | 0.699 | 0.453 | 0.733 | 0.676 | 0.811 | 0.401 | 0.836 | 0.794 |
| RCD-EIRS | 0.722 | 0.431 | 0.738 | 0.718 | 0.702 | 0.452 | 0.738 | 0.689 | 0.809 | 0.411 | 0.829 | 0.800 |
| RCD-contrast | 0.728 | 0.427 | 0.737 | 0.720 | 0.710 | 0.438 | 0.723 | 0.704 | 0.814 | 0.404 | 0.830 | 0.807 |
| RCD-$r_q$ | 0.732 | 0.421 | 0.742 | 0.728 | 0.722 | 0.439 | 0.747 | 0.709 | 0.824 | 0.393 | 0.836 | 0.812 |
| RCD-$r_q r_{G_q}$ | 0.732 | 0.418 | 0.743 | 0.728 | 0.723 | 0.436 | 0.746 | 0.712 | 0.825 | 0.394 | 0.837 | 0.814 |
| SCD* | 0.721 | 0.436 | 0.738 | 0.706 | 0.715 | 0.436 | 0.752 | 0.705 | 0.786 | 0.423 | 0.823 | 0.713 |
| SCD | 0.724 | 0.439 | 0.734 | 0.721 | 0.732 | 0.429 | 0.751 | 0.705 | 0.809 | 0.367 | 0.826 | 0.793 |
| SCD*-EIRS | 0.726 | 0.438 | 0.741 | 0.707 | 0.720 | 0.435 | 0.748 | 0.711 | 0.772 | 0.413 | 0.812 | 0.711 |
| SCD*-$r_q$ | 0.735 | 0.423 | 0.741 | 0.730 | 0.769 | 0.415 | 0.787 | 0.729 | 0.811 | 0.392 | 0.823 | 0.807 |
| SCD*-$r_q r_{G_q}$ | 0.739 | 0.422 | 0.741 | 0.737 | 0.771 | 0.412 | 0.776 | 0.765 | 0.813 | 0.389 | 0.822 | 0.810 |

**Table 5**
Comparison of different operation function.

| Datasets | ASSIST | | | Mooper | | | Junyi | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | ACC | RMSE | ACC-T | ACC | RMSE | ACC-T | ACC | RMSE | ACC-T |
| NCD-add | 0.730 | 0.431 | 0.725 | 0.710 | 0.422 | 0.702 | 0.790 | 0.411 | 0.771 |
| NCD-had | 0.727 | 0.431 | 0.722 | 0.706 | 0.426 | 0.698 | 0.787 | 0.423 | 0.769 |
| NCD-NN | 0.732 | 0.429 | 0.728 | 0.709 | 0.421 | 0.702 | 0.794 | 0.411 | 0.778 |
| KSCD-add | 0.722 | 0.427 | 0.713 | 0.747 | 0.430 | 0.726 | 0.824 | 0.399 | 0.805 |
| KSCD-had | 0.716 | 0.428 | 0.709 | 0.744 | 0.432 | 0.723 | 0.821 | 0.399 | 0.801 |
| KSCD-NN | 0.728 | 0.419 | 0.720 | 0.752 | 0.425 | 0.734 | 0.827 | 0.391 | 0.807 |
| RCD-add | 0.731 | 0.419 | 0.727 | 0.721 | 0.440 | 0.711 | 0.821 | 0.396 | 0.812 |
| RCD-had | 0.728 | 0.419 | 0.721 | 0.718 | 0.442 | 0.702 | 0.819 | 0.398 | 0.805 |
| RCD-NN | 0.732 | 0.418 | 0.728 | 0.723 | 0.436 | 0.712 | 0.825 | 0.394 | 0.814 |
| SCD*-add | 0.737 | 0.425 | 0.734 | 0.767 | 0.413 | 0.764 | 0.811 | 0.391 | 0.809 |
| SCD*-had | 0.728 | 0.434 | 0.723 | 0.764 | 0.419 | 0.719 | 0.809 | 0.392 | 0.807 |
| SCD*-NN | 0.739 | 0.422 | 0.737 | 0.771 | 0.412 | 0.765 | 0.819 | 0.389 | 0.810 |

Liu et al. (Yao et al., 2023), while -contrast represents the Graph Contrast Learning strategy employed in SCD.

1) Our correlation representation exhibits superior performance in addressing the long-tailed distribution compared to the other two strategies. Specifically, on the ASSIST dataset, the ACC-T of -EIRS and -contrast increase by an average of 0.8 % and 2.4 %, respectively, while our -$r_q r_{G_q}$ is improved by 4.5 %. EIRS and contrast eliminate the sparsity of tail nodes by generating potential records or sampling head node records in the view of cognitive state, while our method learns the correlation representation in both cognitive and learning mode views, which can enhance the representation of tail students with more information and eliminate the sparsity of records.

2) Our correlation representation achieves the improvement of tail students, not at the expense of head students. EIRS witnesses an improvement of [0.1 %, 2.3 %] in ACC-T, but witnesses an average drop of 1.2 %. Contrast witnesses an average improvement of 2.4 % in ACC-T but witnesses an average drop of 0.7 %. As for our -$r_q r_{G_q}$, ACC-T is improved by 4.5 % on average, while ACC-H is improved in 6 tasks and is decreased in the other 6 tasks, comparable with baselines on average. The records generated in EIRS are insufficient and noisy, and the contrast learning in SCD drops some interactions of the head students, leading to the decrease of head students. However, our method enhances the tail students in representation level and it does not perform the undersampling for head students, thus the performance of head students is relatively stable.

3) As for RCD and SCD methods based on GNN, both cognitive correlation and the correlation of the learning mode are beneficial for the performance of CD. Compared to RCD, RCD-$r_q$ and RCD-$r_q r_{G_q}$ demonstrate an average improvement of 2.5 % and 2.7 % in ACC-T across the three datasets, respectively. This suggests that both node-level cognitive correlation and graph-level correlation of learning model contribute to the enhancement of representation.

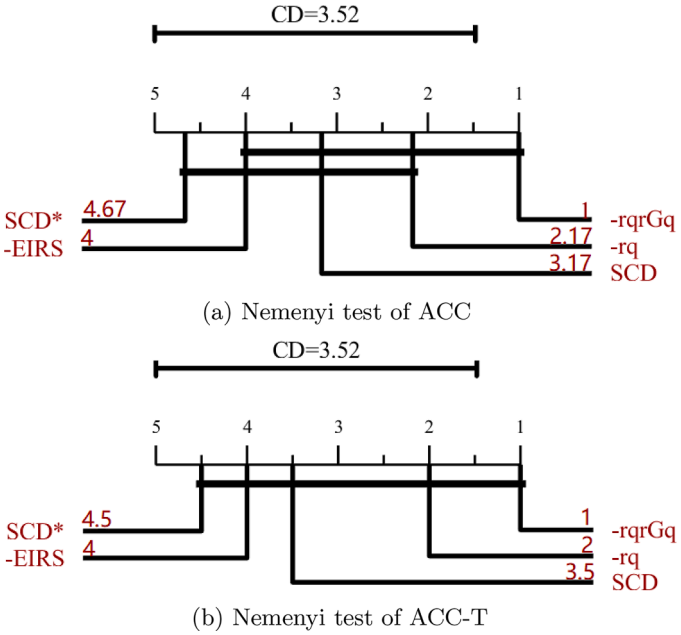
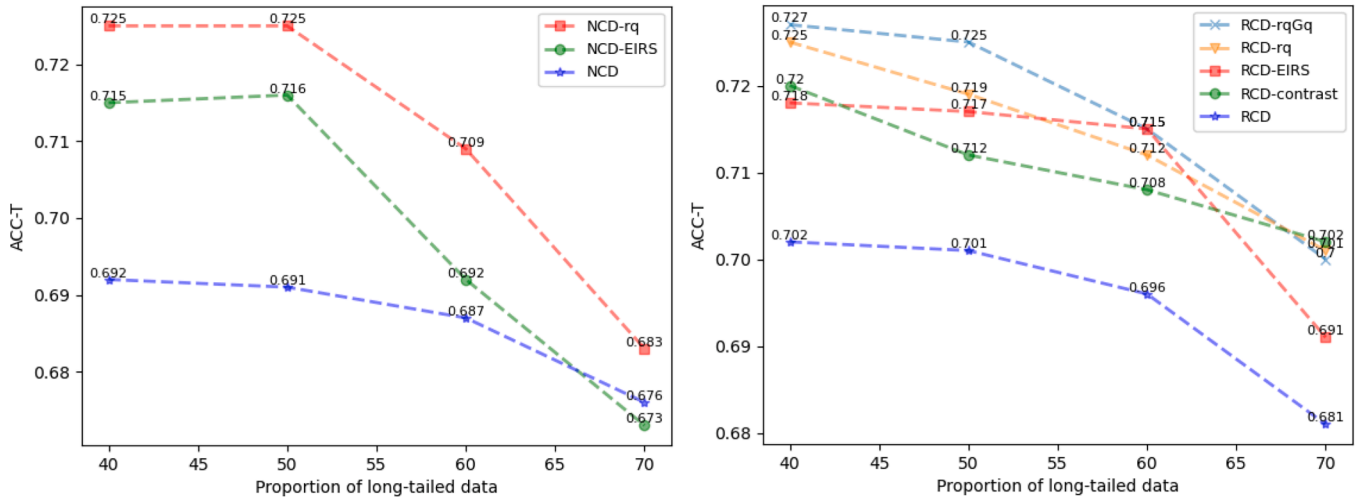
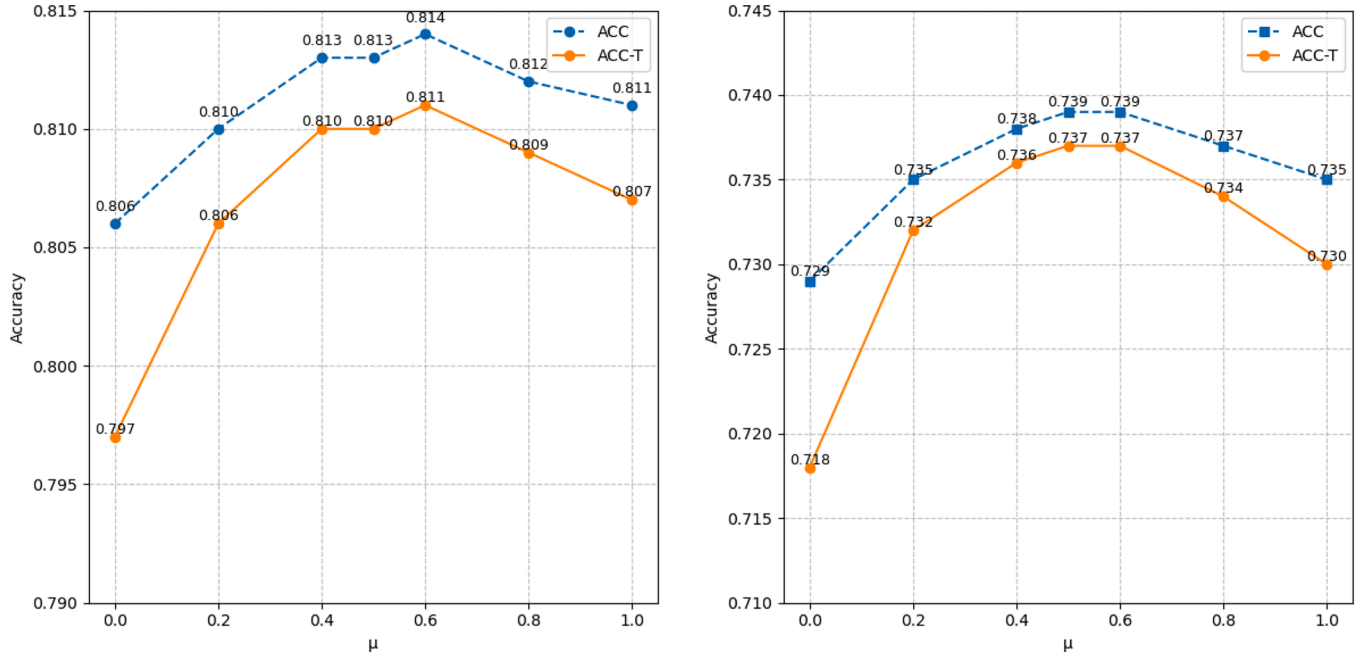



**Fig. 5.** Nemenyi test for three long-tailed mechanisms based on SCD method.




(a) Methods based on NCD (b) Methods based on RCD

**Fig. 6.** The changing of diagnostic accuracy with the proportion of tail students on Junyi dataset.




(a) Junyi dataset. (b) ASSIST dataset.

**Fig. 7.** Sensitiveness of $\mu$ based on SCD-$r_q r_{G_q}$.

In order to better evaluate the effect of each method on long-tailed distribution, we conduct the Nemenyi (Jiang, Cukic, & Menzies, 2008) test for five methods on three datasets with ACC and ACC-T, as shown in Fig. 5. In the results of the Nemenyi test, the smaller the rank number of the model, the better the model. It can be found that -$r_q r_{G_q}$ is significantly better than SCD and -EIRS, which illustrates the outperformance of our correlation representation for long-tailed distribution compared to EIRS and the contrast mechanism in SCD.

#### 4.4.2. *Influence of the operation of enhancement*

In Eq. (9), a second interaction is used to perform the enhancement of the representation. There are several operations used commonly, such as addition, Hadamard product, and a NN with fully connected layer. We compare three operations in Table 5. It can be seen that neural network operation achieves the best performance. It may be because the fully connected layer can model the fusion process between correlation representation and tail representation in a more complex way. Compared with NN, the add and Hadamard product tend to obtain the value of each dimension only based on the corresponding dimension of two vectors, ignoring the complex interaction among different dimensions.

### 4.5. *Influence of tail node proportion*

Fig. 6 illustrates the performance change with the proportion of tail students based on NCD and RCD, in which the proportion of tail students increases from 50 % to 70 % by reducing the records of head students randomly. It can be seen that the diagnosis performance of each method tends to decrease with the increase in the proportion of tail students, and the performance of our method decreases slowly than the other methods. This is because all methods address long-tailed problem with the head students similar to tail students. When the proportion of tail students increases, that of head students decreases, and the information of

head students will be sparse, which will influence the enhancement of tail students in turn. EIRS will predict noisy potential records with insufficient information from the head students. Our method may find the dissimilar head students to learn the correlation representation, which may be insufficient and inaccurate.

### 4.6. The sensitiveness of parameter

The parameter $\mu$ is a trade-off parameter for cognitive correlation and the correlation of the learning model, and the learning correlation is only learned from the graph. Thus, we conduct experiments based on SCD in Junyi and ASSIST, as shown in Fig. 7. It can be seen that the optimal range of $\mu$ falls into [0.5-0.6]. That means the cognitive correlation and the learning model correlation both contribute to the diagnosis of long-tail students.

## 5. Conclusion

The proposed cognitive diagnosis method addresses the challenge of the long-tailed distribution of response records. With proposed correlation representation, the representation of tail students iss enhanced, addressing the diagnosis limitations posed by sparsity of records. In the near future, our aim is to further explore the influence factors of students selecting exercises and design a more efficient correlation model between head and tail students. Moreover, the long-tailed distribution of the relation between exercises and knowledge concepts also requires further research.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Xuegang Hu reports financial support was provided by National Natural Science Foundation of China. Yuhong Zhang reports financial support was provided by Natural Science Foundation of Anhui Province. Yuhong Zhang reports financial support was provided by University Synergy Innovation Program of Anhui Province. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Yuhong Zhang:** Conceptualization, Methodology; **Tiancheng He:** Data curation, Software, Writing - review & editing; **Shengyu Xu:** Visualization, Software; **Mian Wang:** Writing - original draft; **Chenyang Bu:** Formal analysis, Writing - review & editing; **Xuegang Hu:** Supervision.

## Acknowledgement

This work is supported by the National Natural Science Foundation of China (under grant 62076085), the Natural Science Foundation of Anhui Province (under grant Natural Science Foundation of Anhui Province2208085MF170) and the University Synergy Innovation Program of Anhui Province (GXXT-2022-040).

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